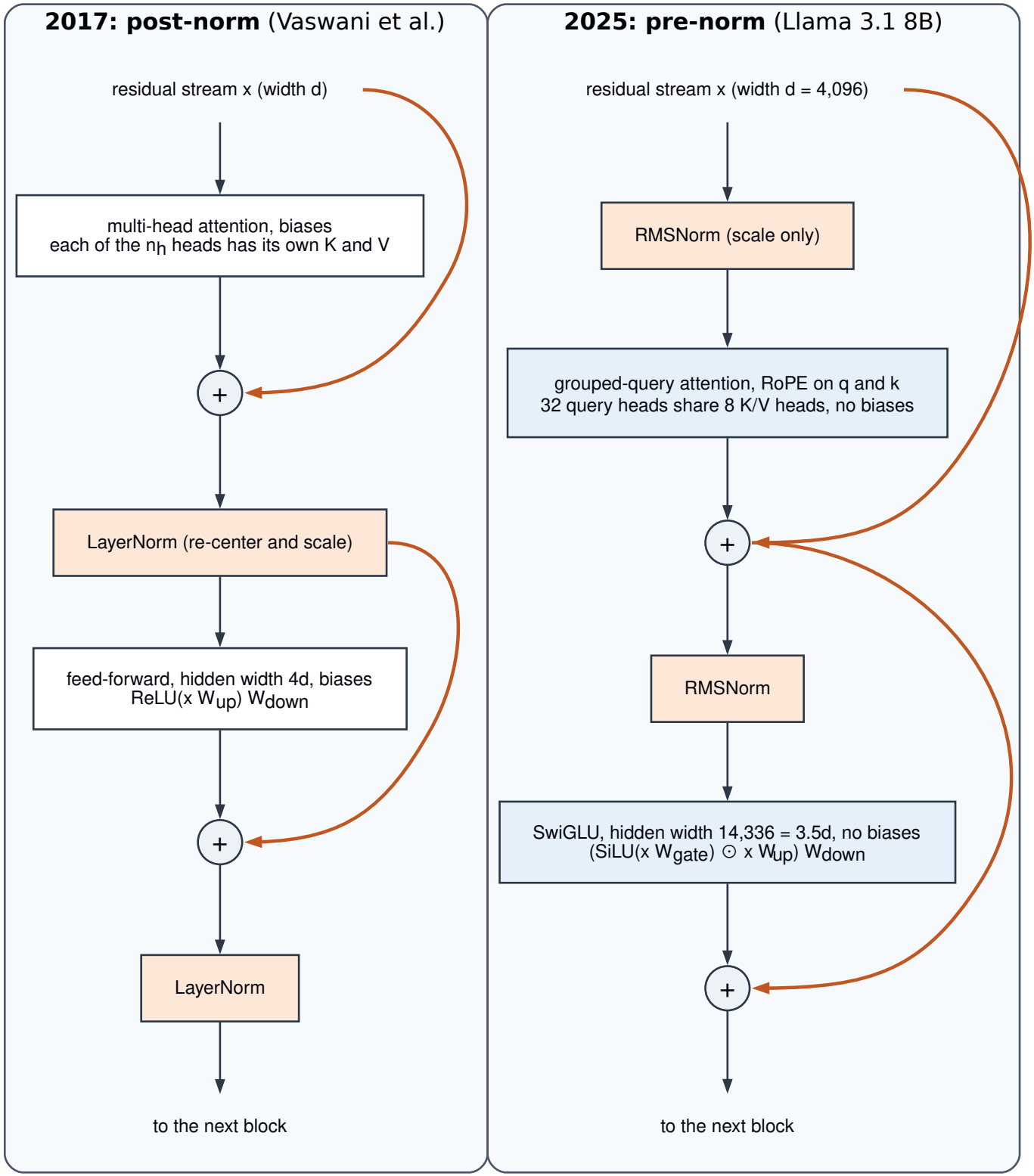
2017: post-norm (Vaswani et al.)
residual stream x (width d)
multi-head attention, biases
each of the n_h heads has its own K and V
+
LayerNorm (re-center and scale)
feed-forward, hidden width 4d, biases
ReLU(x W_{up}) W_{down}
+
LayerNorm
to the next block
2025: pre-norm (Llama 3.1 8B)
residual stream x (width d = 4,096)
RMSNorm (scale only)
grouped-query attention, RoPE on q and k
32 query heads share 8 K/V heads, no biases
+
RMSNorm
SwiGLU, hidden width 14,336 = 3.5d, no biases
(SiLU(x W_{gate}) ⊙ x W_{up}) W_{down}
+
to the next block